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Type Conversion in Python:

Type conversion is used to convert values from one data type to another, either automatically by the Python interpreter or manually by the programmer using in built functions. Note that not all data types can be converted into each other. For example, there's no way a string consisting of English letter alphabets can be converted into an integer.

Suppose a situation where a person is traveling to America from India and that person wants to buy something from a store in America. But that person only has cash in INR(Indian Rupees). So the person goes to get their money converted from INR to USD (American Dollar) from a local bank in America.

Data Type conversion examples

<code>int(10)</code>	<code>=> 10</code>	<code>float(10)</code>	<code>=> 10.0</code>
<code>int(10.1)</code>	<code>=> 10</code>	<code>float(10.1)</code>	<code>=> 10.1</code>
<code>int('10')</code>	<code>=> 10</code>	<code>float('10')</code>	<code>=> 10.0</code>
<code>int('10.1')</code>	<code>=> Error</code>	<code>float('10.1')</code>	<code>=> 10.1</code>
<code>int('kitten')</code>	<code>=> Error</code>	<code>float('kitten')</code>	<code>=> Error</code>
<code>str(10)</code>	<code>=> '10'</code>		
<code>str(10.1)</code>	<code>=> '10.1'</code>		
<code>str('kitten')</code>	<code>=> 'kitten'</code>		

Types of Type Conversion in Python

There are two types of type conversion in Python:



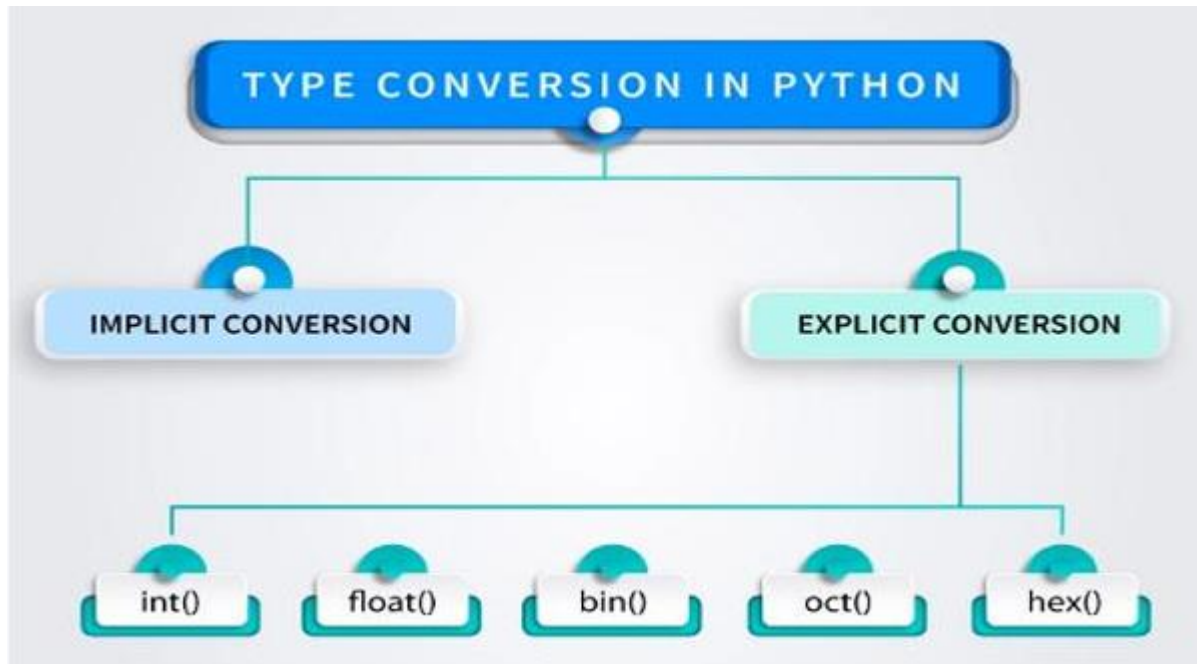
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Implicit Type Conversion in Python

1. Implicit data type conversion is done automatically by the python compiler without any involvement of the user.
2. The compiler converts smaller data types into larger data types to prevent any loss of data, for example, conversion of int to float.
3. Implicit type conversion will occur only if the data types used in the statement are compatible. If the data types are incompatible, then a type error will be raised.

Example - Implicit Type Conversion of int to float:

num1 = 2

num2 = 12.2



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```
res = num1 + num2

print("Data type of num1: ", type(num1))

print("Data type of num2: ", type(num2))

print("Data type of res: ", type(res))

print("Value of res: ", res)
```

Output

Data type of num1: <class 'int'>

Data type of num2: <class 'float'>

Data type of res: <class 'float'>

Value of res: 14.2

Explicit Type Conversion in Python

- Explicit Type Conversion is also known as Type Casting in Python.
- Explicit type conversion is done by the user whereas Implicit Type Conversion is done by a Python interpreter.

```
# assigning string to x
```

```
sample_str = "0010101"
```

```
num = 85
```

```
#type casting
```



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```
conv_str = float(sample_str)
```

```
conv_num = float(num)
```

```
print("String converted into floating-point: ",conv_str)
```

```
print("Integer converted into floating-point: ",conv_num)
```

Output:

String converted into floating-point: 10101.0

Integer converted into floating-point: 85.0



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